

SERGIU IOAN MOCAN

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TECHNICAL SKILLS

Languages: C#, Java, C++, Python, TypeScript

Frameworks: .NET, Spring Boot, React, Node.js, Avalonia UI

Cloud & DevOps: AWS (ECS, ECR, S3, RDS, VPC), AWS CDK (IaC), Docker, Kubernetes, Azure DevOps CI/CD, Bash

Data: SQL Server, PostgreSQL, Redis

EXPERIENCE

TSS Yonder | *Software Engineer*

2025 – Present

Core Stack: .NET (C#), Spring Boot (Java), React (TypeScript), AWS (CDK, ECS, EventBridge, SQS), SQL Server, Azure DevOps

Migrating a Dealer Management System used by 750+ automotive businesses across Western Europe from a legacy VB6 desktop app to a cloud-native AWS platform.

- Led the cloud migration to a React/TypeScript frontend and a polyglot microservices backend, provisioning AWS infrastructure with CDK and deploying to ECS Fargate. Scaled system to reliably handle 23,000+ daily transactions and 1,000+ concurrent users with zero downtime.
- Cut deployment time 86% (90 → <12 min) by building end-to-end Azure DevOps pipelines covering automated testing, SQL migrations, and multi-environment AWS rollouts.
- Architected an event-driven communication layer across microservices using Amazon EventBridge and SQS, with idempotent consumer workers guaranteeing once processing and fault tolerance during peak traffic.
- Ended recurring data divergence by co-engineered a REST sync layer that keeps an AWS RDS source of truth consistent with two on-premises SQL Server instances.

Accounting System Development S.R.L. | *Software Engineer*

2022 – 2024

Core Stack: Python (FastAPI), React (TypeScript), SQLAlchemy, PostgreSQL

- Collaborated with an agile engineering team to build an internal web platform that automated end-to-end document workflows (ingestion, generation, routing, reporting), replacing manual daily procedures.
- Shipped filtering and statistics tooling, giving the client real-time visibility into document volume and processing status.

PROJECTS

Blockchain Digital Voting System | *Solidity, Ethereum, TypeScript, React*

- Architected a 9-contract system behind EIP-1967 upgradeable proxies for safe in-place upgrades without state migration.
- Implemented O(1) vote deduplication with bitwise arrays, cutting on-chain storage by 96%.
- Built an O(log n) Merkle proof pipeline featuring off-chain generation, on-chain eligibility verification, and a React dApp issuing proof-carrying transactions with RBAC-driven UI.

AirCursor | *Python, OpenCV, MediaPipe, NumPy, PyAutoGUI*

- Engineered a 30 FPS webcam gaze pipeline tracking 468 facial and 10 iris landmarks to translate head pose and eye gaze into OS cursor motion.
- Formulated dual-vector pose compensation using OpenCV solvePnP and relative iris ratios to isolate true gaze from head rotation.
- Applied adaptive 1€ low-pass filtering, dwell-time hovering, and blink aspect-ratio detection for jitter-free cursor control and clicking.

EDUCATION

Babeş-Bolyai University | M.Sc., Distributed Systems in the Internet

Babeş-Bolyai University | B.Sc., Computer Science (English Section)

Teaching: Private tutor for the UBB CS admission exam (algorithms, data structures, logic)

CERTIFICATIONS

AWS Certified Solutions Architect – Associate (SAA-C03)

In Progress